

Heart Disease Prediction using ML through IoT enabled ECG Monitor

Team Name: Heartify

Team Members: Adrija Ghosh, Shyantana Kundu, Apratim Haldar & Suvojit Maity

Department: Computer Science & Engineering, Electronics & Communication
Engineering

College: Institute Of Engineering & Management, IEM Kolkata

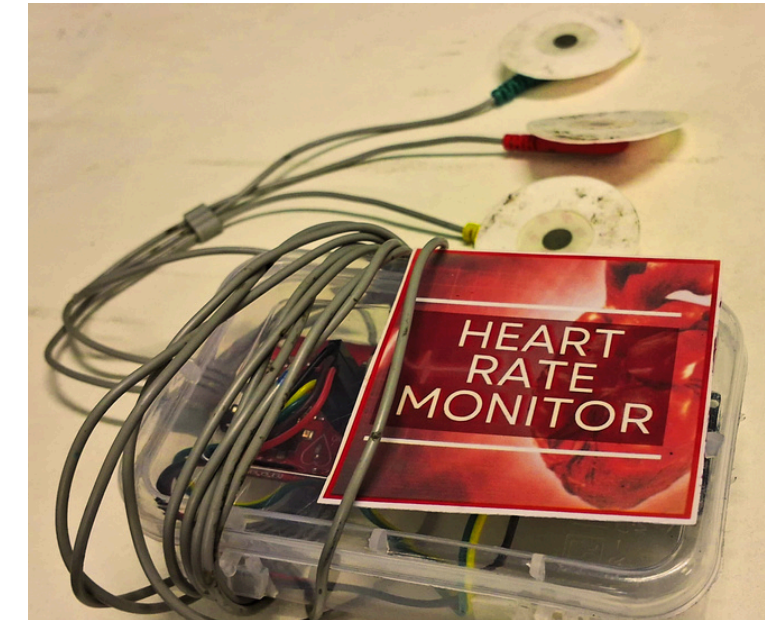
Introduction

➤ The **heart disease prediction system** combines a **hardware** device with **software website** to monitor heart health.

➤ The device includes an **AD8232 sensor**, **Arduino Uno**, and **ESP32 module** to continuously track heart rate .

➤ Heart rate data is sent to the backend via **Flask**, which stores it in **MongoDB**.

➤ A **Random Forest machine learning model** in **Python** analyzes the data to predict heart disease risk .



Heart Rate Monitor Device

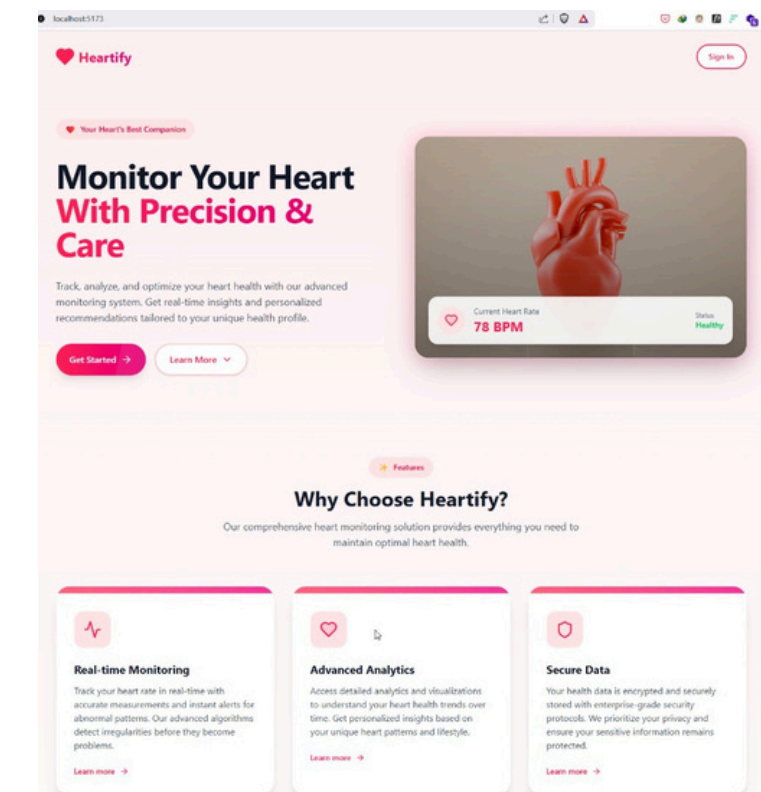
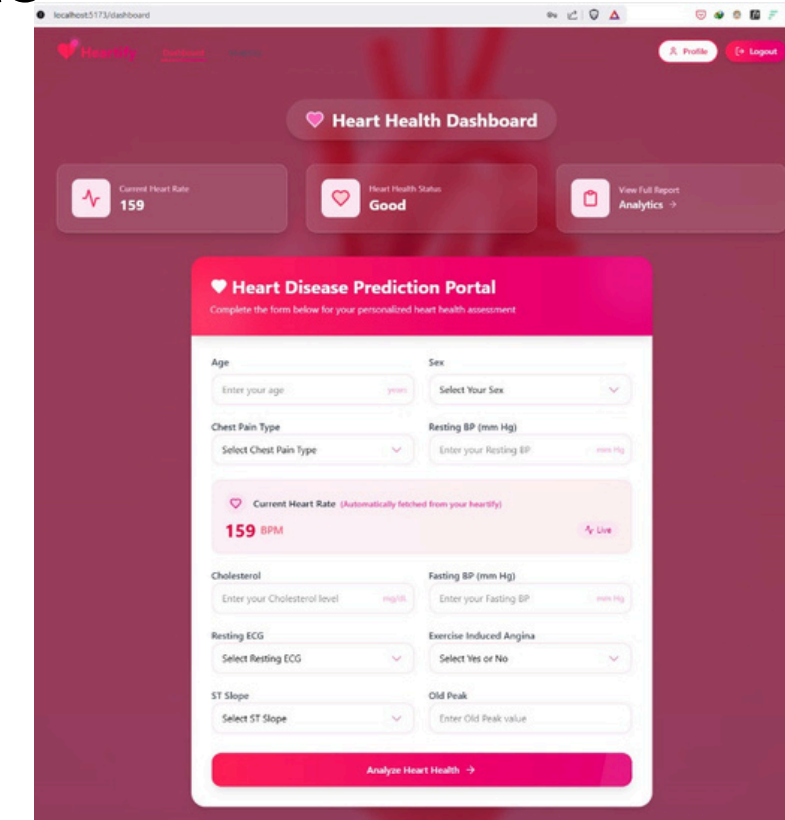
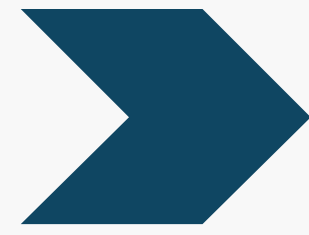
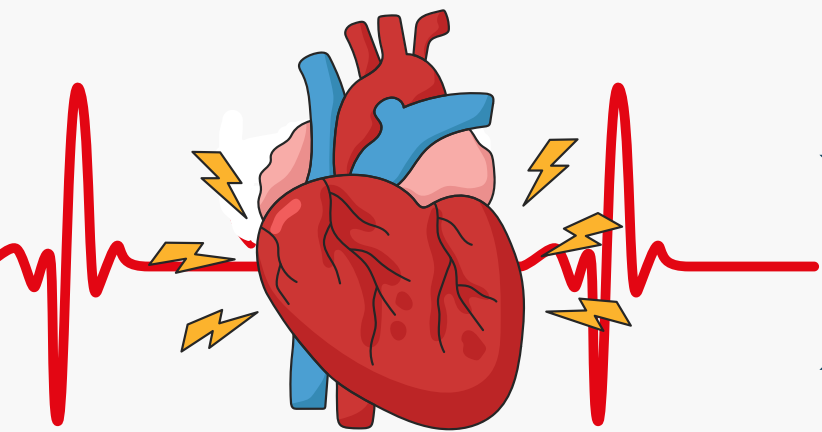
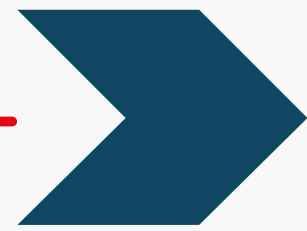


Fig.1. Screenshots of website

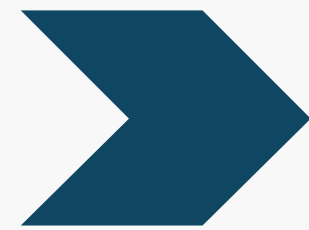
Problem Statement



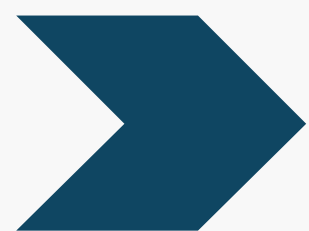
Already existing heart rate monitoring devices are so **much expensive** and complex in use.



Existing solutions often **lack integration** between hardware and software, making them complex for the user.



In most cases, patients **cannot check** their heart rate during the **onset of a problem**, leaving it undetected.



Early detection and **continuous monitoring** of heart health is necessary for reducing the risk of heart diseases before they become critical.

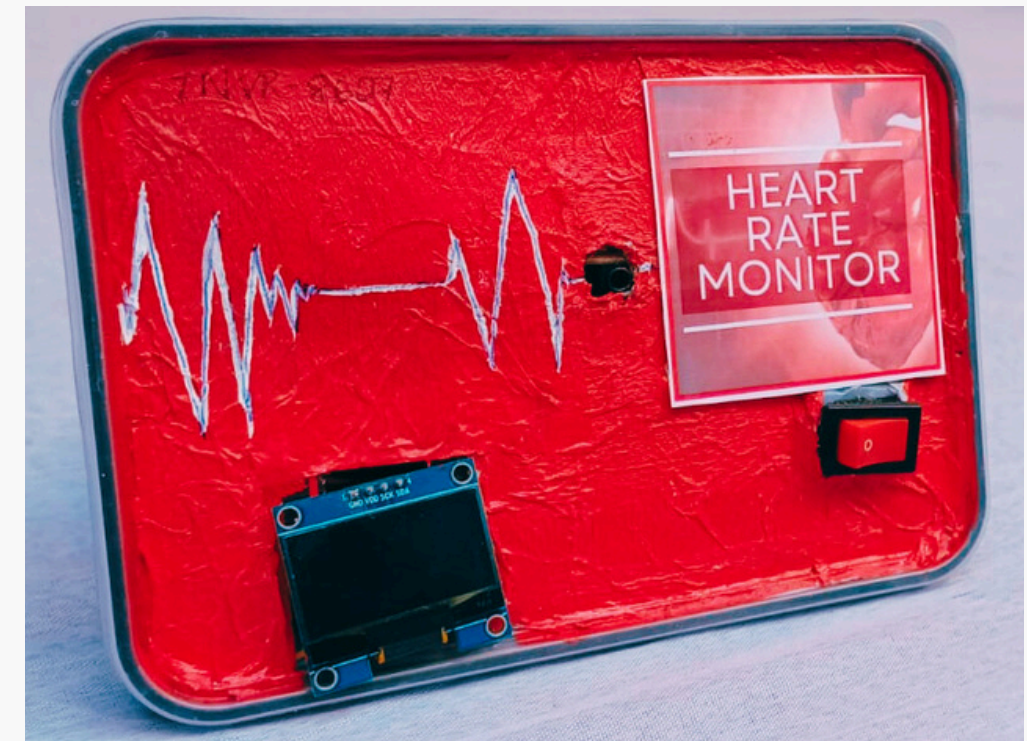


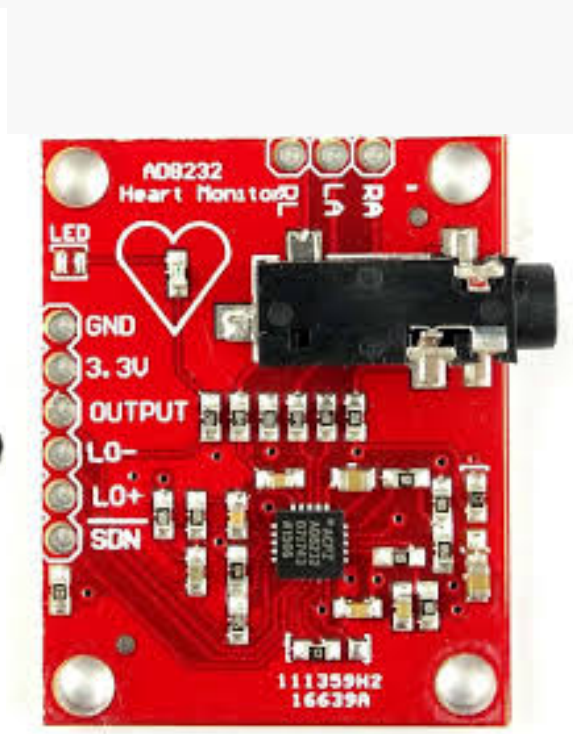
Fig. 2. First Prototype of our Device

Tools and Technology Used



Hardware Tools

Software Tools



ESP 32

AD 8232 Sensor



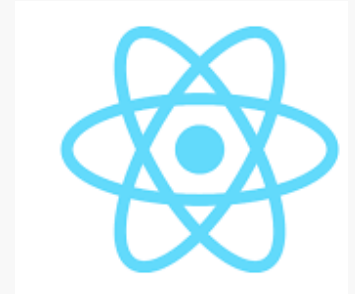
Python



Flask Library



MongoDB



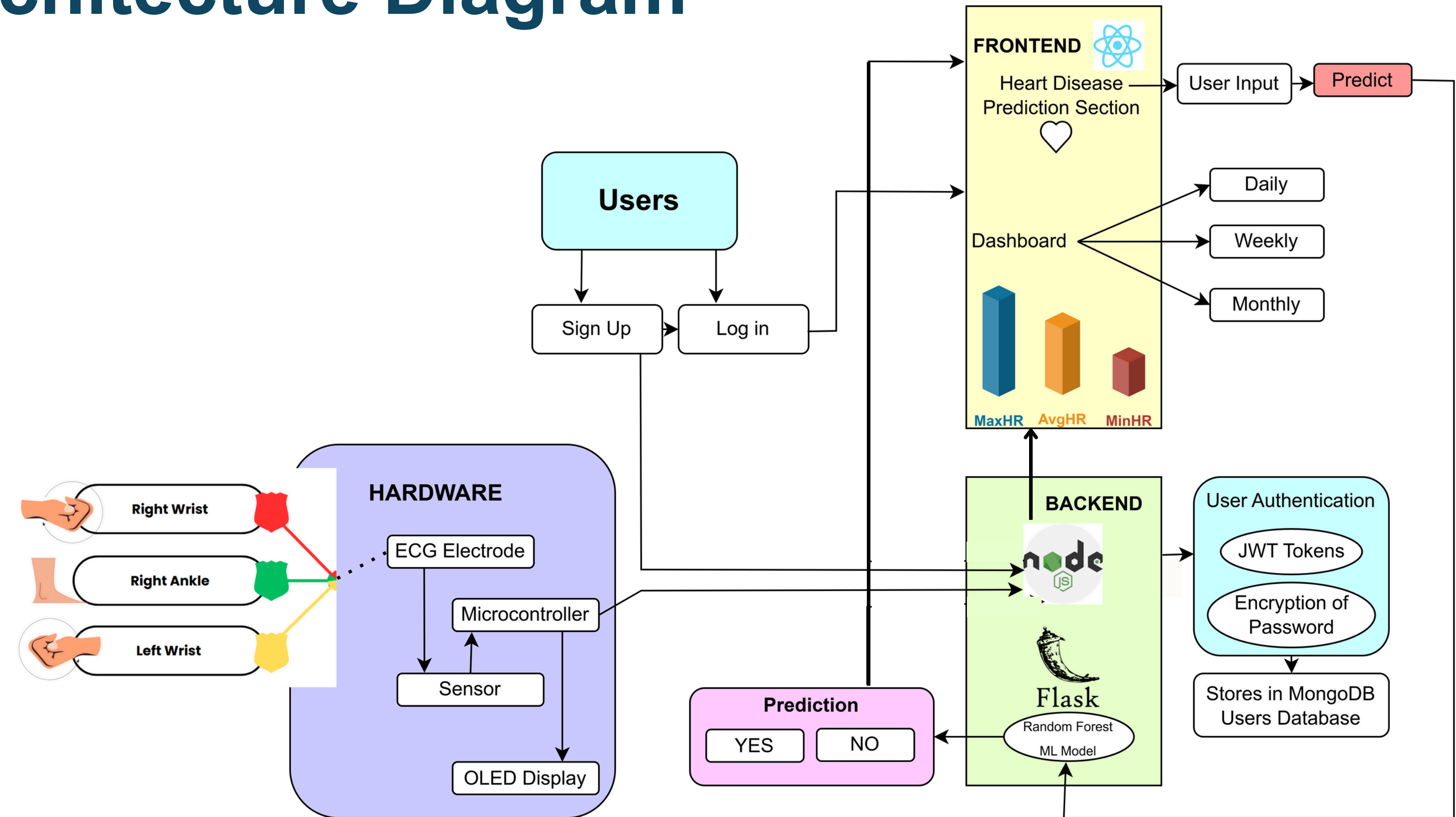
React.js

[Github Repo](#)

[HRM Video Link](#)



Architecture Diagram



Modules of the project



Hardware (Heart Rate Monitor Device)

Our machine uses **ECG electrodes** to detect real-time human heart data, calculates heart rate and sends in to **MongoDB** by through flask server .

Database (MongoDB)

A database is connected in backend to **authenticate** the login/signup process. It also fetches live heart data from the hardware machine to the website.

Website

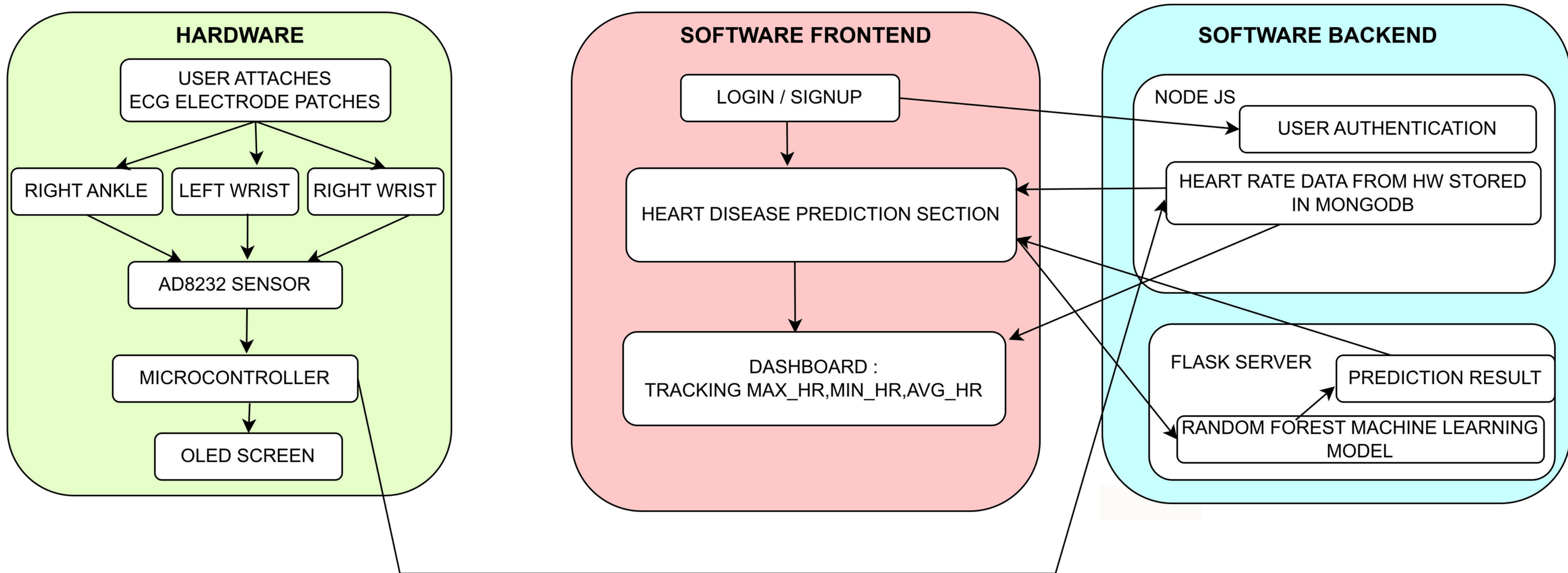
It has two sections, **Heart Disease Prediction** section and **Dashboard**. In prediction section, a **pretrained machine learning model** is deployed that predicts if a person is at risk of heart disease or not and the dashboard displays the person's daily, weekly and monthly **trends of heart rate**.

Machine Learning Model

At the backend of website a pre-trained unique ML model is deployed with almost **88% accuracy** to predict heart disease .



Working Flow



Model Building and Result Analysis

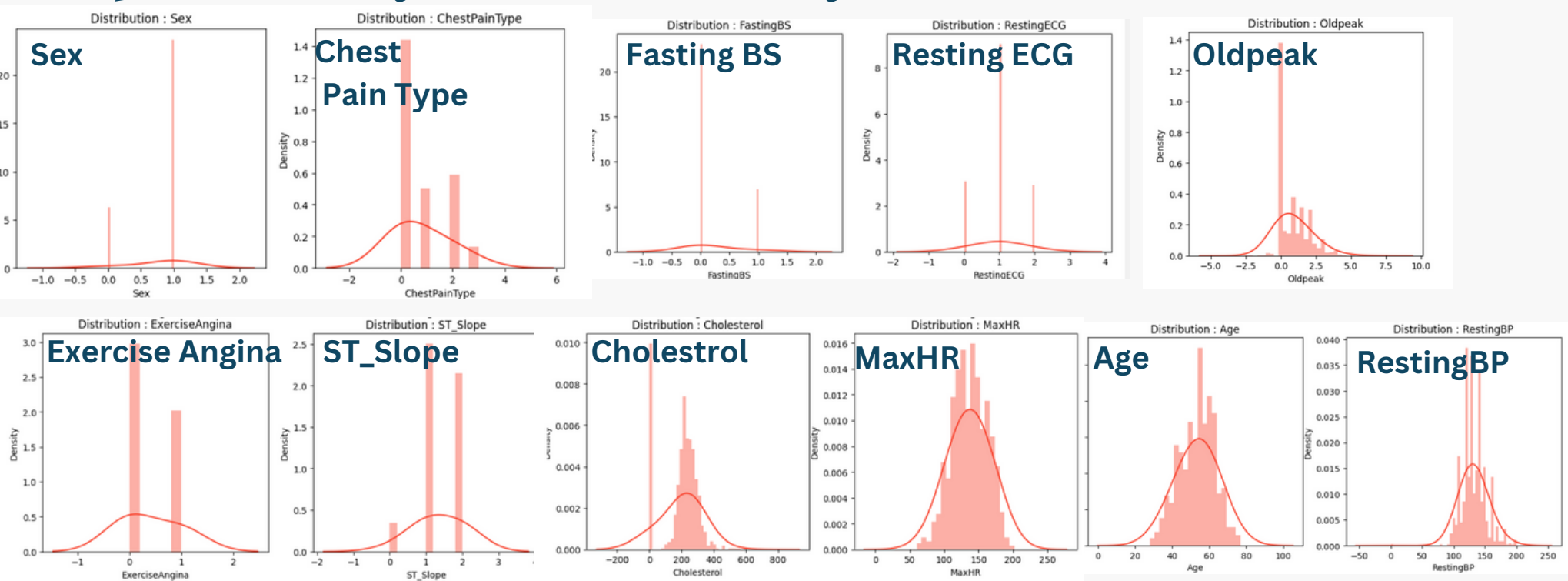
Machine Learning Model

We have created a novel model that **uses features** of:

- Maximum Heart Rate (measured)
- Type of Chest Pain
- Resting Blood Pressure
- Cholestrol level
- Age
- Presence of fasting blood sugar
- Resting ECG value
- Pain after exercise
- Old peak
- ST slope
- Gender

to predict whether a person is at a risk of heart disease or not.

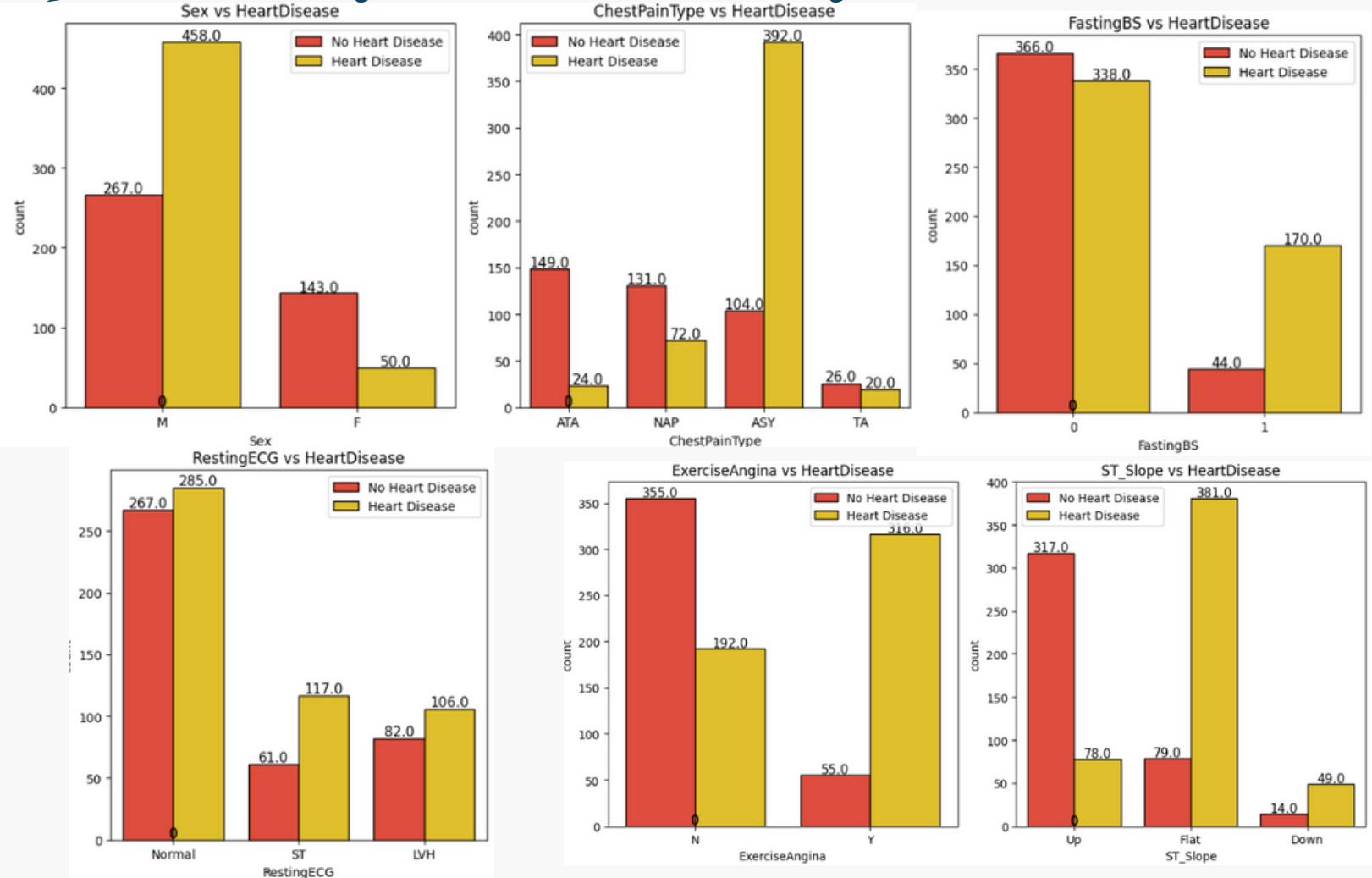
Exploratory Data Analysis (Univariate)



Conclusion

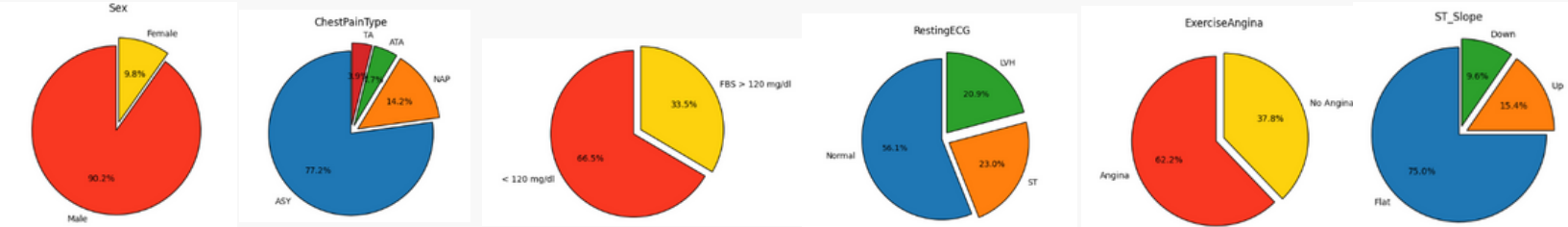
On analysis, it is observed that the data is normally distributed.

Exploratory Data Analysis (Bivariate)



Observation

- Males are more prone to heart disease.
- ASY type of chest pain implies heart disease.
- If exercise induced angina is present, high chances of heart disease.
- Flat ST Slope implies high tendency of heart disease.



Model Building and Result Analysis

The model was built using a Random Forest Classifier to predict heart disease, leveraging a dataset with both categorical and numerical features.

Random Forest is chosen over other models as it has both **low bias and low variance** which gives better performance.

```
def train_and_evaluate_model(X_train, y_train, X_test, y_test):  
  
    rf_model = RandomForestClassifier(random_state=42)  
    rf_model.fit(X_train, y_train)  
    y_pred = rf_model.predict(X_test)
```

Accuracy: 0.875

Precision: 0.8962264150943396

Recall: 0.8878504672897196

F1 Score: 0.892018779342723

Classification Report:

	precision	recall	f1-score	support
0	0.85	0.86	0.85	77
1	0.90	0.89	0.89	107
accuracy			0.88	184
macro avg	0.87	0.87	0.87	184
weighted avg	0.88	0.88	0.88	184

The model has **87.5% accuracy** with precision of **90%**. The model successfully identifies **89%** of all true heart disease cases. F1 Score (Class 1) of 89% represents a balance between precision and recall for heart disease predictions .

Overfitting Analysis

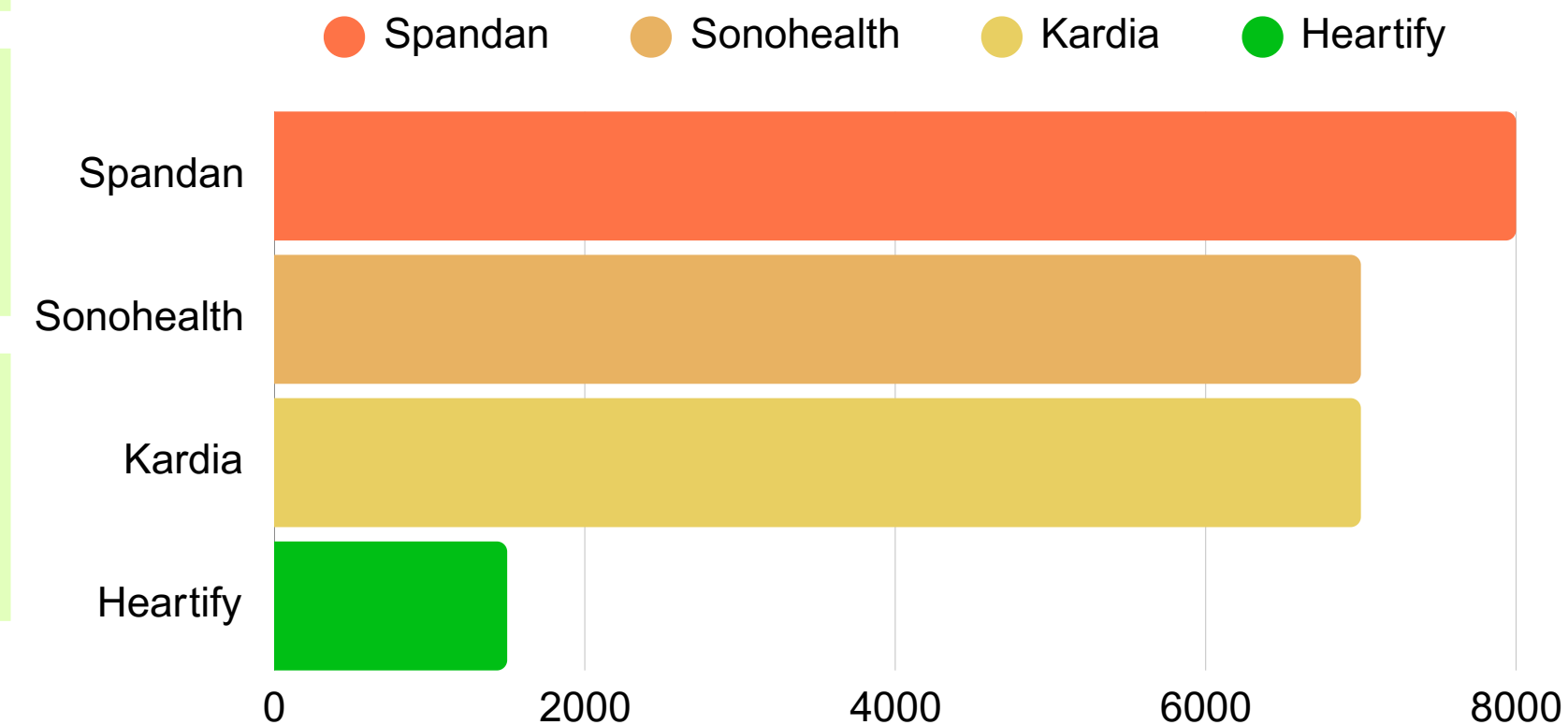
- The F1 scores for both classes are close to each other, suggesting that the model is not overfitting and is effectively balancing the trade-off between precision and recall.
- Both precision and recall scores are close, indicating the model generalizes well and is not biased toward any specific class.

Existing Solutions

Metric	Sunfox Spandan	Sono health	Kardia	Heartify
Price	₹8000- ₹20,000	₹7000+	₹7000+	₹1500
ML model	Used	Not Used	Not Used	Used
Website linking	Not done	Not done	Not done	Done
Available in India	Yes	No	No	Yes

What makes us better?

- Integrated hardware and software system.
- Auto-fetching heart data into ML prediction section.
- Cheaper than competitors.
- More accurate heart results for daily use.



Achievements



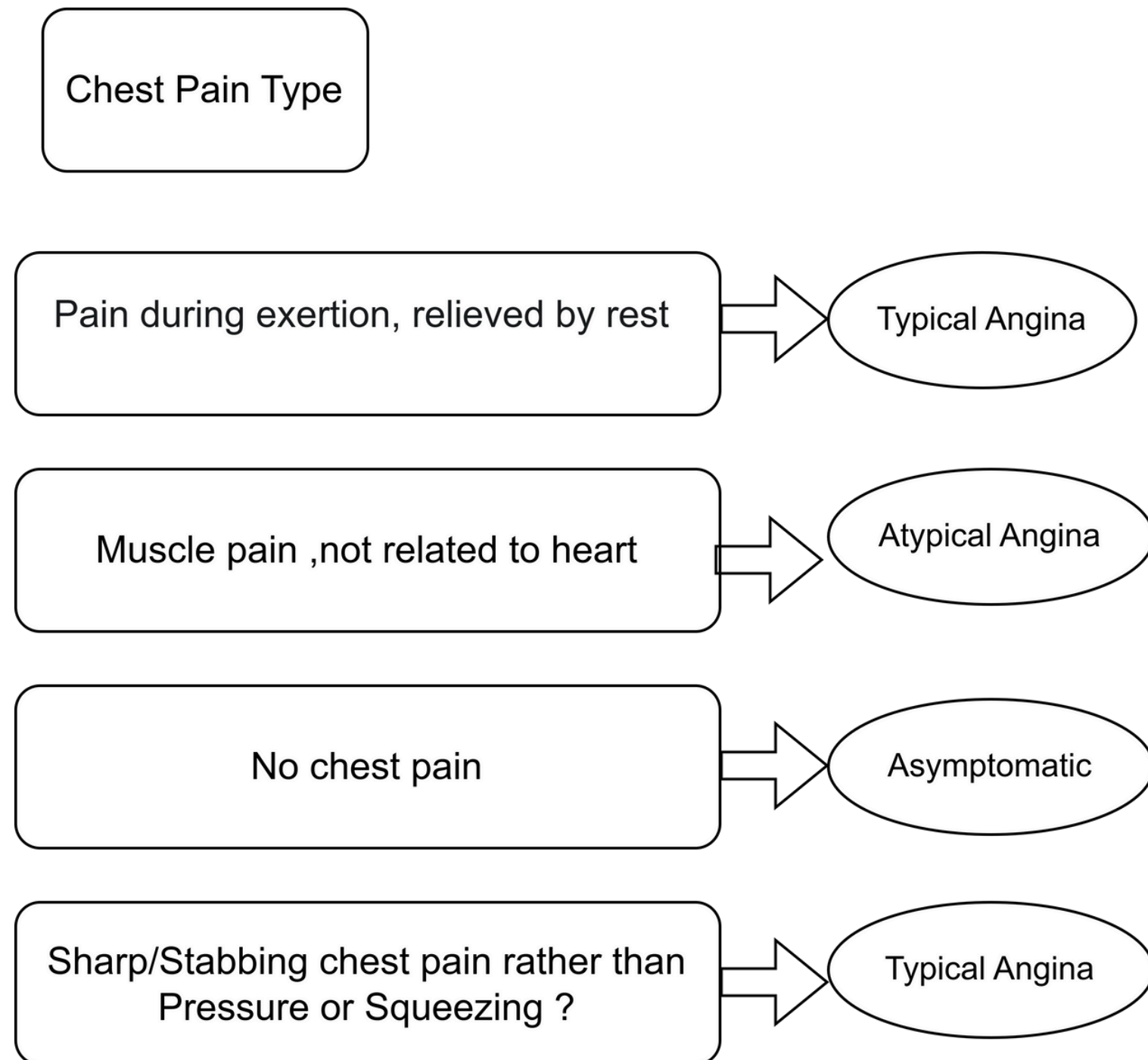
2nd Runners Up – Nation level HACKATHON on “ Frontier Applications of Big data Analytics in Healthcare Informatics”

Led Team Heartify to secure 3rd place for developing an ML-based Predictive ECG Measuring Device using Random Forest and XGBoost, integrating low-level system design and hardware-software sync. Proudly represented IEM Kolkata in health tech innovation!



Future Work

Question Answer based Input from Users



Similarly for OldPeak and ST Slope questions, We can ask users to upload their ECG report and using **CNN** we can get values of OldPeak and ST Slope.

If all the values is not present to the users, the required feild from the model will be removed. In that case the model will take most common values through **missing data imputation**, however in that case the output will not be accurate.

Conclusion

- Successfully developed an **integrated hardware-software** solution for real-time heart rate monitoring.
- Deployed a **machine learning model** on the website to predict heart disease risk.
- Established a **live database** for heart rate tracking with daily, weekly, and monthly trends.
- Created an **affordable and accessible** healthcare tool with potential for large-scale impact.
- Moving towards **clinical validation**, regulatory approvals, and further product refinement.
- With continuous improvements, the solution is on track to become a **market-ready product**.

